



Modelling Financial Markets

Seminar 9: Realised Volatility — Hands-on



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CLT: Central Limit Theorem
- ▣ DM: Diebold–Mariano (test)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ IQ: Integrated Quarticity
- ▣ JB: Jarque–Bera (test)
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKElihood (loss function)

Acronyms used in this material (2/2)

- ▣ RF: Random Forest
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SHAR: Semivariance HAR model
- ▣ TQ: Tripower Quarticity
- ▣ TSRV: Two-Scale Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

Outline

Introduction

Part A: Derivations

Part B: Computation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the moments and the feasible CLT of RV, the noise bias and the robustness of BV to jumps
- ▣ Test jumps on many days with FDR control and a periodicity correction
- ▣ Test HAR as a restricted AR(22), explain HARQ by errors in variables
- ▣ Compare variance forecasts with QLIKE, Diebold–Mariano, Giacomini–White and the model confidence set

- ▣ Background: Lecture 9; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** nine computational exercises with inference on SPY and Bitcoin (B1–B9)
- ▣ **Part C:** an open question, a seed for the team project (C1), the critique of an AI answer (C2) and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Units: returns in %, daily variances in $\%^2$; annualise with 252 days (SPY) or 365 days (Bitcoin)
- Every daily realised measure is itself an estimate: report how precise it is

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (moments of RV, feasible CLT), A3 (noise bias, optimal sampling), A5 (HAR as a restricted AR(22)), A6 (robustness of QLIKE)
- Part B: B1 (jump test on SPY), B3 (sampling frequency with block bootstrap), B5 (log-HAR with Newey–West errors)

Proposed: you solve them

- Part A: A2 (BV under a jump; model: A1), A4 (two-scale RV and its rate; model: A3), A7 (attenuation and HARQ; model: A5), A8 (joint Wald test in MZ; model: A6)
- Part B: B2 (Bitcoin jumps; model: B1), B4 (standardised returns; model: B3), B6 (5-day forecasts; model: B5, A8), B7 (HAR-CJ; model: B5, B1), B8 (roughness; model: B3), B9 (model confidence set; model: B5, B6)
- Part C: C1, open question and project seed; C2, critique of an AI answer; C3, machine learning vs HAR on SPY (lecture case study; model: B5, A6)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter9_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `arch`, `matplotlib`
- SPY, 5-minute bars of the regular US session, October 2020 – September 2026; Bitcoin, 5-minute bars, 24-hour UTC days, September 2024 – September 2026
- Daily closes of SPY (adjusted) and Bitcoin for GARCH; SPY total daily variance = intraday RV + squared overnight return

Formulas Used Today

- $RV_t = \sum_{i=1}^M r_{t,i}^2$; $BV_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}|$; $se(RV_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- Jump test [Huang & Tauchen \(2005\)](#): $z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(\mathbf{1}, \mathbf{TQ}_t/BV_t^2)}}$, $\theta = \pi^2/4 + \pi - 5$
- Noise: $\mathbb{E}[RV^{(n)} | \sigma] = IV + 2n\omega^2$; $n^* = (IQ/(4\omega^4))^{1/3}$ [Bandi & Russell \(2008\)](#); two-scale $RV = RV^{\text{avg}} - \frac{\bar{n}}{n} RV^{\text{all}}$ [Zhang, Mykland & Aït-Sahalia \(2005\)](#)
- log-HAR [Corsi \(2009\)](#): $\ln RV_{t+1} = \beta_0 + \beta_d \ln RV_t + \beta_w \overline{\ln RV_t^{(5)}} + \beta_m \overline{\ln RV_t^{(22)}} + \varepsilon_{t+1}$
- QLIKE = $RV/F - \ln(RV/F) - 1$ [Patton \(2011\)](#); DM: $t = \bar{d}/se_{\text{NW}}(\bar{d})$ [Diebold & Mariano \(1995\)](#); MZ: $RV_t = a + bF_t + u_t$, $H_0 : a = 0, b = 1$

A1: Moments of RV and the Feasible CLT [Solved]

Task

- ▣ Context: one trading day is split into M intervals of length $\Delta = 1/M$; volatility is constant, so the intraday returns are $r_i = \sigma\sqrt{\Delta} Z_i$ with Z_i i.i.d. $N(0,1)$ and the daily variance is $IV = \sigma^2$.
- ▣ (a) Derive the mean and the variance of $RV = \sum_{i=1}^M r_i^2$, using $\mathbb{E}Z^2 = 1$ and $\text{Var}(Z^2) = 2$.
- ▣ (b) Show that realised quarticity $RQ = \frac{M}{3} \sum_i r_i^4$ is an unbiased estimator of $IQ = \sigma^4$, using $\mathbb{E}Z^4 = 3$.
- ▣ (c) Apply the delta method to obtain the standard errors of $\sqrt{RV}$ and of $\ln RV$, and write the 95% confidence interval for IV built on the log scale.
- ▣ (d) Compute the relative width of this interval for $M = 78$ (5-minute returns in a 6.5-hour session).
- ▣ Report: the four formulas, the CI factor for $M = 78$, and one sentence on why the same CI remains valid when σ changes during the day.

A1: Moments of RV and the Feasible CLT – Solution [Solved]

Solution

- (a) Mean: $\mathbb{E}[\text{RV}] = \sum_i \sigma^2 \Delta \mathbb{E}Z_i^2 = M\sigma^2/M = \sigma^2$, so RV is unbiased for IV.
- (a) Variance: the terms are independent, so $\text{Var}(\text{RV}) = M\sigma^4\Delta^2 \text{Var}(Z^2) = M \cdot \sigma^4/M^2 \cdot 2 = 2\sigma^4/M$.
- (b) $\mathbb{E}[r_i^4] = \sigma^4\Delta^2\mathbb{E}Z^4 = 3\sigma^4/M^2$, hence $\mathbb{E}[\text{RQ}] = \frac{M}{3} \cdot M \cdot 3\sigma^4/M^2 = \sigma^4 = \text{IQ}$; therefore $\widehat{\text{Var}}(\text{RV}) = 2 \text{RQ}/M = \frac{2}{3} \sum_i r_i^4$.
- (c) With $g(x) = \sqrt{x}$: $\text{se}(\sqrt{\text{RV}}) \approx \text{se}(\text{RV})/(2\sqrt{\text{RV}})$; with $g(x) = \ln x$: $\text{se}(\ln \text{RV}) \approx \sqrt{2 \text{RQ}/M}/\text{RV}$.
- (c) The 95% CI is $\text{RV} \exp(\pm 1.96 \text{se}(\ln \text{RV}))$: it is always positive and asymmetric in levels.
- (d) With constant volatility $\text{RQ}/\text{RV}^2 \approx 1$, so $\text{se}(\ln \text{RV}) \approx \sqrt{2/78} = 0.160$ and the CI factor is $e^{\pm 0.314} = [0.73, 1.37]$.
- Interpretation: one day of 5-minute data pins IV down to roughly $-27\%/+37\%$; with stochastic σ the studentised limit still holds because the convergence is stable (Lecture 9), so RQ may replace the random IQ.

A2: Bipower Variation under a Jump [Proposed]

Solution discussed in class

Task

- Context: bipower variation $BV = \frac{\pi}{2} \frac{M}{M-1} \sum_{i=2}^M |r_i| |r_{i-1}|$ and the ratio statistic $z = \frac{(RV - BV)/RV}{\sqrt{\theta M^{-1} \max(1, TQ/BV^2)}}$,
 $\theta = \pi^2/4 + \pi - 5 = 0.609$ [Huang & Tauchen \(2005\)](#).
- (a) Suppose one jump of size κ falls in an interior interval j ($1 < j < M$), so $r_j = \kappa + r_j^c$; show that, to leading order, it enters BV only through the two products $|r_{j-1}| |\kappa| + |\kappa| |r_{j+1}|$ (one product if $j = 1$ or $j = M$), that these are $O_p(\Delta^{1/2})$, and conclude that $BV \rightarrow IV$ while $RV \rightarrow IV + \kappa^2$.
- (b) For the six 5-minute returns (%) 0.08, -0.15, 0.03, 0.10, -0.04, -0.90 (the last one is a news jump), compute RV, BV, the jump part $RV - BV$ and its share of RV.
- (c) Compute z with $M = 6$ and the given $TQ = 0.0097$, applying the floor $\max(1, TQ/BV^2)$, and decide at the 0.1% level (critical value 3.09).
- (d) Derive the largest value z can take with $M = 6$ and explain what it says about the power of the test on such a short window.
- Report: the derivation in (a), a small table of the numbers in (b)–(c), the bound in (d) and one sentence of interpretation. Model: A1.

A3: Noise Bias and the Optimal Frequency [Solved]

Task

- ▣ Context: observed log prices are $p_i^{\text{obs}} = p_i^* + \varepsilon_i$, with ε_i i.i.d. $(0, \omega^2)$ and independent of the efficient price p^* ; a day contains n returns.
- ▣ (a) Write the observed return in terms of r_i^* and the noise, and derive $\mathbb{E}[\text{RV}^{(n)}] = \text{IV} + 2n\omega^2$ and $\text{cov}(r_i, r_{i-1}) = -\omega^2$.
- ▣ (b) Take the approximation $\text{MSE}(n) \approx 2\text{IQ}/n + 4n^2\omega^4$ [Bandi & Russell \(2008\)](#) and derive the MSE-optimal number of returns n^* from the first-order condition.
- ▣ (c) Propose two consistent estimators of ω^2 and state the limit in which each is consistent.
- ▣ (d) With $\text{IV} = 1\%^2$, $\text{IQ} = 1$, $\omega = 0.01\%$ and a 23400-second session, compute $\mathbb{E}[\text{RV}]$ at 5 minutes, 1 minute and 1 second, and n^* with the implied sampling interval.
- ▣ Report: the derivations, the three expectations, n^* , and which interval you would use in practice and why.

A3: Noise Bias and the Optimal Frequency – Solution [Solved] (1/2)

Solution

- (a) $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$; given the path of σ (zero drift) the cross terms have mean 0, so $\mathbb{E}[r_i^2 \mid \sigma] = \mathbb{E}[r_i^{*2} \mid \sigma] + 2\omega^2$; summing over n returns gives $\mathbb{E}[\text{RV}^{(n)} \mid \sigma] = IV + 2n\omega^2$.
- (a) Adjacent returns share only ε_{i-1} (with opposite signs), so $\text{cov}(r_i, r_{i-1}) = \text{cov}(-\varepsilon_{i-1}, \varepsilon_{i-1}) = -\omega^2$.
- (b) $\frac{d}{dn} \text{MSE} = -2IQ/n^2 + 8n\omega^4 = 0$ gives $n^3 = IQ/(4\omega^4)$, i.e. $n^* = (IQ/(4\omega^4))^{1/3}$; the second derivative is positive, so this is a minimum.
- (c) $\hat{\omega}^2 = \text{RV}^{\text{all}}/(2n)$: $\mathbb{E}[\hat{\omega}^2 \mid \sigma] = \omega^2 + IV/(2n)$, and a law of large numbers for the noise terms gives $\hat{\omega}^2 \xrightarrow{P} \omega^2$ as $n \rightarrow \infty$; alternatively $-\widehat{\text{cov}}(r_i, r_{i-1})$, consistent as the number of returns grows.

A3: Noise Bias and the Optimal Frequency – Solution [Solved] (2/2)

Solution

- (d) $\mathbb{E}[\text{RV}] = 1 + 2 \cdot 78 \cdot 0.0001 = 1.0156$ at 5 minutes, 1.0780 at 1 minute ($n = 390$) and 5.6800 at 1 second ($n = 23\,400$): at one second RV is almost six times IV.
- (d) $n^* = (1/(4 \times 10^{-8}))^{1/3} = 292$, i.e. one return every $23\,400/292 = 80$ seconds.
- Interpretation: in this model 1 minute is close to the optimum (MSE 0.0112 vs 0.0103 at n^*), while 5 minutes has 2.5 times the minimal MSE (0.0259).
- The usual 5-minute choice rests on an argument outside this model: ω is uncertain and real noise is not i.i.d., and 5-minute RV is hard to beat empirically [Liu, Patton & Sheppard \(2015\)](#).

A4: Two-Scale RV and Its Rate [Proposed]

Solution discussed in class

Task

- Context: a less liquid BVB stock with $IV = 1\%^2$ and noise standard deviation $\omega = 0.03\%$ (wider spread, larger tick); on one day, 1-minute returns ($n = 390$) give $RV^{\text{all}} = 1.70$, and the average over $K = 5$ shifted 5-minute grids is $RV^{\text{avg}} = 1.10$.
- (a) Compute the noise bias $2n\omega^2$ at 5 and at 1 minute, and the optimal n^* of A3 with $IQ = 1$ and its sampling interval in seconds.
- (b) Compute two-scale RV, $TSRV = RV^{\text{avg}} - \frac{\bar{n}}{n}RV^{\text{all}}$ with $\bar{n} = (n - K + 1)/K$, with and without the small-sample factor $1/(1 - \bar{n}/n)$, and estimate ω^2 from RV^{all} .
- (c) Show algebraically that TSRV removes the noise bias to first order.
- (d) Explain heuristically why the error of TSRV is of order $n^{-1/6}$ rather than $n^{-1/2}$, by balancing the two variance terms in K .
- Report: the numbers in (a)–(b), the two derivations and one sentence on when TSRV is worth using. Model: A3.

A5: HAR as a Restricted AR(22) [Solved]

Task

- Context: log-HAR [Corsi \(2009\)](#), $y_{t+1} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + \varepsilon_{t+1}$, with $y_t = \ln RV_t$, $\bar{y}_t^{(5)}$ the mean of $y_t, \dots, y_{t-4}$ and $\bar{y}_t^{(22)}$ the mean of $y_t, \dots, y_{t-21}$.
- (a) Write the implied coefficients $\phi_1, \dots, \phi_{22}$ of the equivalent AR(22) model.
- (b) Write the restriction matrix R such that HAR is $R\phi = 0$ in the AR(22) with intercept, and count the restrictions.
- (c) Write the Wald statistic with a Newey–West (HAC) covariance and give its limiting distribution under HAR.
- (d) For the rounded SPY estimates $\hat{\beta} = (-0.08, 0.36, 0.34, 0.17)$, compute $\phi_1, \phi_{2..5}, \phi_{6..22}$ and the persistence $\sum_k \phi_k$.
- Report: the three formulas, the matrix structure, the numbers in (d) and what the persistence implies for shocks.

A5: HAR as a Restricted AR(22) – Solution [Solved]

Solution

- (a) Lag 1 appears in all three components: $\phi_1 = \beta_d + \beta_w/5 + \beta_m/22$; lags 2–5 in the weekly and monthly ones: $\phi_k = \beta_w/5 + \beta_m/22$; lags 6–22 only in the monthly one: $\phi_k = \beta_m/22$.
- (b) Rows $e_k - e_{k+1}$ for $k = 2, 3, 4$ (3 rows: $\phi_2 = \dots = \phi_5$) and $k = 6, \dots, 21$ (16 rows: $\phi_6 = \dots = \phi_{22}$); with the zero intercept column R is 19×23 : 22 lags minus 3 free slopes.
- (c) $W = (R\hat{\phi})^\top (R\hat{V}_{\text{NW}}R^\top)^{-1} R\hat{\phi} \rightarrow \chi^2_{19}$; HAC because the errors of the log-RV regression are heteroskedastic and possibly autocorrelated.
- (d) $\phi_1 = 0.36 + 0.34/5 + 0.17/22 = 0.436$; $\phi_{2..5} = 0.34/5 + 0.17/22 = 0.076$; $\phi_{6..22} = 0.17/22 = 0.008$;
 $\sum \phi_k = \beta_d + \beta_w + \beta_m = 0.87$.
- Interpretation: a persistence of 0.87 means shocks to log RV decay slowly but the process is stationary; the data test of the 19 restrictions is B5(d).

A6: Why QLIKE Is Robust and MSE on Volatility Is Not [Solved]

Task

- ▣ Context: the forecast F is evaluated against the proxy RV_t , which is conditionally unbiased:
 $\mathbb{E}[RV_t | \mathcal{F}_{t-1}] = \mathbb{E}[IV_t | \mathcal{F}_{t-1}]$ [Patton \(2011\)](#); $QLIKE(RV, F) = RV/F - \ln(RV/F) - 1$.
- ▣ (a) Minimise $\mathbb{E}[QLIKE(RV_t, F) | \mathcal{F}_{t-1}]$ over F and show that the minimiser is $\mathbb{E}[IV_t | \mathcal{F}_{t-1}]$, the same as with the true variance.
- ▣ (b) Minimise $\mathbb{E}(\sqrt{RV} - \sqrt{F})^2$ (MSE on volatility) over F and use Jensen's inequality to compare the minimiser with $\mathbb{E}RV$.
- ▣ (c) With constant IV and $RV = IV \chi_M^2/M$, compute $F^*/IV = (\mathbb{E}\sqrt{RV})^2/IV$ for $M = 1$ (squared daily return), $M = 6$ and $M = 78$.
- ▣ (d) With true variance 1, compute the QLIKE of the forecasts $F = 0.5$ and $F = 2.0$ and compare with their MSE.
- ▣ Report: the two first-order conditions, the three ratios, the two losses and which loss you would use to rank variance forecasts.

A6: Why QLIKE Is Robust and MSE on Volatility Is Not – Solution [Solved]

Solution

- (a) $\partial_F \mathbb{E}[\text{RV}/F - \ln \text{RV} + \ln F - 1] = -\mathbb{E}\text{RV}/F^2 + 1/F = 0$ gives $F = \mathbb{E}[\text{RV} \mid \mathcal{F}_{t-1}] = \mathbb{E}[\text{IV}_t \mid \mathcal{F}_{t-1}]$; the second derivative $2\mathbb{E}\text{RV}/F^3 - 1/F^2 = 1/F^2 > 0$ there.
- (b) $\mathbb{E}(\sqrt{\text{RV}} - \sqrt{F})^2 = \mathbb{E}\text{RV} - 2\sqrt{F} \mathbb{E}\sqrt{\text{RV}} + F$, minimal at $\sqrt{F} = \mathbb{E}\sqrt{\text{RV}}$; by Jensen $(\mathbb{E}\sqrt{\text{RV}})^2 < \mathbb{E}\text{RV}$, so the optimal forecast is biased down.
- (c) $F^*/\text{IV} = \frac{2}{M} (\Gamma(\frac{M+1}{2})/\Gamma(\frac{M}{2}))^2 = 0.637$ for $M = 1$, 0.920 for $M = 6$, 0.994 for $M = 78$.
- (d) QLIKE: $F = 0.5$ gives $2 - \ln 2 - 1 = 0.307$, $F = 2.0$ gives $0.5 + \ln 2 - 1 = 0.193$; MSE is 0.25 and 1, the opposite ranking.
- Interpretation: against squared daily returns MSE on volatility rewards a forecast 36% too low; QLIKE keeps the ranking of the true variance and penalises under-prediction, which matters for risk.

A7: Attenuation in HAR and the Case for HARQ [Proposed]

Solution discussed in class

Task

- Context: $RV_t = IV_t + u_t$ with $\mathbb{E}[u_t | \sigma] = 0$ and $\text{Var}(u_t | \sigma) = 2IQ_t/M$ (the asymptotic CLT approximation of A1); measurement errors are serially uncorrelated and uncorrelated with IV and with the innovations η (zero cross-moments; full independence of IV would contradict a variance that depends on IQ_t); the latent variance follows $IV_{t+1} = \beta_0 + \beta IV_t + \eta_{t+1}$ with $\beta > 0$.
- (a) Derive the probability limit of the OLS slope in the regression of RV_{t+1} on RV_t and show that it is smaller than β .
- (b) Conditional on IQ_t , and under the working assumption that IV_t has mean μ and variance σ_{IV}^2 not depending on IQ_t , derive the best linear weight on RV_t for predicting IV_t ; show that it falls with IQ_t and that this motivates the HARQ slope $\beta_d + \beta_Q \sqrt{RQ_t}$ with $\beta_Q < 0$ Bollerslev, Patton & Quaedvlieg (2016).
- (c) Propose an instrumental-variables estimator that removes the attenuation, name the instrument and state the assumptions for its validity and relevance.
- Report: the plim formula, the conditional weight, the IV estimator, and a sentence on what the SPY estimate of β_Q in B5(e) says. Model: A5.

A8: Joint Wald Test in the Mincer–Zarnowitz Regression [Proposed]

Solution discussed in class

Task

- ▣ Context: Lecture 9, SPY one-day forecasts, January 2022 – September 2026; loss difference $d_t = \text{QLIKE}(\log\text{-HAR}) - \text{QLIKE}(\text{GARCH})$ has mean $\bar{d} = -0.0327$ and Newey–West standard error 0.0142.
- ▣ The MZ regression $\text{RV}_t = a + bF_t + u_t$ for GARCH gives $\hat{a} = 0.2418$ (SE 0.2310), $\hat{b} = 0.7046$ (SE 0.2382), with estimated correlation $\text{corr}(\hat{a}, \hat{b}) = -0.9750$ (HAC covariance); keep all four decimals, since with a correlation near -1 the Wald statistic is sensitive to rounding.
- ▣ (a) Compute the Diebold–Mariano statistic and its two-sided p -value, and decide at 5%.
- ▣ (b) Compute the separate t -statistics for $a = 0$ and for $b = 1$ and decide at 5%.
- ▣ (c) Build the covariance matrix of $(\hat{a}, \hat{b})$ from the standard errors and the correlation, compute the joint Wald statistic for $(a, b) = (0, 1)$ and its χ^2_2 p -value.
- ▣ (d) Sketch the 95% confidence ellipse and explain when separate and joint tests can disagree. Report the three statistics, the decisions and their interpretation. Model: A6.

B1: Jumps in SPY [Solved] (1/2)

Task

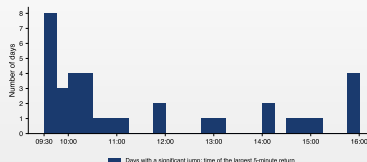
- Data: SPY 5-minute bars of the regular US session, `data/market/intraday/SPY.US_5m.csv`, October 2020 – September 2026, 1 489 days, 78 returns per full day (42 on half-days), returns $r_{t,i} = 100 \Delta \ln P$ in %.
- Method: the ratio statistic z_t of [Huang & Tauchen \(2005\)](#) (formula sheet) for every day, with one-sided p -values $p_t = 1 - \Phi(z_t)$.
- (a) Count the days that reject at 5%, 1% and 0.1%, and compare each count with $\alpha \times$ the number of days expected under no jumps.
- (b) On the 0.1% jump days, find the 5-minute interval with the largest absolute return and plot the histogram of these times.
- (c) Compute the jump share of total RV, $\sum_t J_t / \sum_t RV_t$, with $J_t = RV_t - BV_t$ on 0.1% jump days and 0 otherwise.

B1: Jumps in SPY [Solved] (2/2)

Task (continued)

- (d) Apply Benjamini–Hochberg at a 5% and a 1% false discovery rate [Benjamini & Hochberg \(1995\)](#), and Bonferroni at 5% ($p_t < 0.05/N$), to the N daily p -values; count the jump days under each (BH controls the FDR for valid p -values that are independent or positively dependent [Benjamini & Yekutieli \(2001\)](#)).
- (e) Divide each return by the intraday periodicity factor f_i (standard-deviation estimator of [Boudt, Croux & Laurent \(2011\)](#)), returns first scaled by $\sqrt{BV_t/M_t}$ and repeat (a) and (d).
- Report: a table of counts (raw, FDR, Bonferroni, with and without periodicity), the histogram, and the jump share.
- Interpretation: are there more rejections than the level alone would produce? Do the jump times point to scheduled news? Which count would you use to build a jump regressor?

B1: Jumps in SPY – Solution [Solved] (1/2)



Solution

- (a) Rejections: 235 at 5% (expected 74.5), 110 at 1% (expected 14.9), 35 at 0.1% (expected 1.5); the mean of z_t is 0.50 and its standard deviation 1.14, instead of 0 and 1.
- (b) Top days: 2021-08-27 ($z = 5.22$, jump 49% of RV, interval ending 10:05), 2022-01-03 ($z = 5.21$, ending 09:50), 2025-08-25 ($z = 4.77$, ending 16:00); the histogram peaks after the open.
- (c) With 0.1% jump days the jump component is 2.8% of total RV.

B1: Jumps in SPY – Solution [Solved] (2/2)

Solution

- (d) FDR 5%: 42 jump days; FDR 1%: 9; Bonferroni 5%: 7.
- (e) With periodicity-corrected returns: 167 rejections at 5%, 20 at 0.1%, and 17 under FDR 5%.
- Reading: the raw counts far exceed αN because of real jumps and because the U-shape makes adjacent returns unequal, which biases BV down; after FDR control and the periodicity correction only 17 days remain.
- Decision: build the jump regressor from FDR-controlled, periodicity-corrected days; the largest remaining ones point to scheduled news, e.g. 2021-08-27 at 10:00–10:05, when the Federal Reserve chair's Jackson Hole speech was released.

B2: Jumps in Bitcoin [Proposed]

Solution discussed in class

Task

- Data: Bitcoin 5-minute bars, `data/market/intraday/BTC-USD.CC_5m.csv`, 24-hour UTC days with at least 95% of the bars, September 2024 – September 2026, 659 days, 288 returns per day.
- (a) Apply the ratio test of B1 at 0.1% to every day, count the jump days and test with a one-sided binomial test whether their share exceeds the level 0.001 (the test treats the daily rejections as independent Bernoulli(0.001) draws under the null).
- (b) Split the days into weekends and weekdays and compare the share of jump days and the mean ratio BV/RV in the two groups.
- (c) Compute the jump share of total variation, as in B1(c).
- Report: counts, binomial p -value, the weekend/weekday table and the jump share.
- Interpretation: why are jump days more frequent than for SPY while jumps carry less variance? What makes a calm weekend day look “jumpy”? Would you model weekends and weekdays together? Model: B1.

B3: Which Sampling Frequency? [Solved] (1/2)

Task

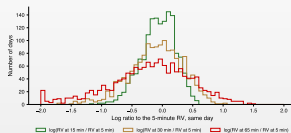
- Data: SPY 5-minute bars of the regular US session, `data/market/intraday/SPY.US_5m.csv`, October 2020 – September 2026, 1 489 days, 78 returns per full day (42 on half-days), returns $r_{t,i} = 100 \Delta \ln P$ in %.
- (a) Compute daily RV from returns every 5, 15, 30 and 65 minutes, both on one grid and as the average over all shifted grids (subsampling), and report the annualised volatility of mean RV, $\sqrt{252 \overline{RV}}$.
- (b) For each frequency, compute the ratio of mean RV to mean 5-minute RV and a 95% CI with a moving-block bootstrap [Künsch \(1989\)](#) (blocks of 20 days, 2 000 resamples).
- (c) Compute the correlation between today's log RV at each frequency and tomorrow's 5-minute log RV.
- (d) Compute the Parzen realised kernel [Barndorff-Nielsen et al. \(2008\)](#) on the 5-minute returns and the ratio of its mean to mean RV, with the same bootstrap; the course data are 5-minute bars, so 1-minute or tick estimators are a project extension.

B3: Which Sampling Frequency? [Solved] (2/2)

Task (continued)

- Report: one table with volatility, ratio [CI] and predictive correlation per frequency, the kernel ratio [CI], and a histogram of $\ln(RV^{(k)}/RV^{(5)})$.
- Interpretation: is there a measurable noise bias at 5 minutes? Why a block bootstrap and not an i.i.d. one? Which frequency gives the most useful measurement for forecasting?

B3: Which Sampling Frequency? – Solution [Solved]



Interval	Vol. (one grid)	Vol. (avg. grids)	Ratio [95% CI]	Corr. next day	Corr. (avg. grids)
5 min	12.93	12.93	1.000 [1.000, 1.000]	0.73	0.73
15 min	12.88	12.79	0.992 [0.954, 1.035]	0.71	0.72
30 min	12.75	12.77	0.973 [0.933, 1.015]	0.70	0.72
65 min	12.76	12.88	0.974 [0.913, 1.037]	0.64	0.70

Solution

- Procedure: build the price grid per day, compute RV on each grid, average across shifted grids, resample 20-day blocks of the paired daily series, recompute the ratio of means in each resample and take the 2.5% and 97.5% percentiles.

B3: Which Sampling Frequency? (reading) – Solution [Solved]

Solution

- (a)–(b) All CIs contain 1, so there is no measurable noise bias for SPY at 5 minutes, consistent with the flat signature plot.
- Block bootstrap: daily RVs are strongly autocorrelated; i.i.d. resampling would destroy this dependence and make the CI too narrow.
- (c) The predictive correlation falls from 0.73 (5 min) to 0.64 (65 min, one grid): sparse RV is noisier; averaging the shifted grids recovers part of it (0.70).
- (d) Kernel / RV = 0.952 with CI [0.910, 0.987] (median bandwidth $H = 7$): the kernel adds back small negative autocorrelations at 5–35 minutes ($\hat{\rho}_1 = -0.008$); significant but economically minor.
- Answer: 5-minute RV is the most useful measurement here; noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid stocks).

B4: Returns Standardised by Realised Volatility [Proposed]

Solution discussed in class

Task

- Data: SPY open-to-close returns (1 489 days, 2020–2026) and Bitcoin 24-hour returns (659 days, 2024–2026), with the same-day RV from 5-minute bars [Andersen et al. \(2001\)](#).
- (a) Compute the kurtosis of $u_t = r_t/\text{sd}(r)$ and of $z_t = r_t/\sqrt{\text{RV}_t}$, each with a moving-block bootstrap 95% CI (blocks of 20 days, 2 000 resamples).
- (b) Run the Jarque–Bera test on z_t and decide at 5%; state that its χ^2_2 reference distribution assumes i.i.d. data.
- (c) Compute the share of $|z_t| > 3$ and compare it with the Normal value 0.27%.
- Report: a table with the kurtosis [CI] of u and z , JB (p) and the tail share for both assets.
- Interpretation: where do the heavy tails of daily returns come from? Why is the kurtosis of z_t slightly below 3? What does this imply for a VaR model (Chapter 7)? Model: B3.

B5: log-HAR for SPY with Robust Inference [Solved] (1/2)

Task

- Data: SPY total daily variance $RV_t + r_{\text{overnight},t}^2$ (5-minute RV plus the squared close-to-open return), 1 467 days after the first month [Corsi \(2009\)](#).
- (a) Estimate log-HAR by OLS on the full sample and report the usual and Newey–West standard errors with 7 lags (rule $\lfloor 4(T/100)^{2/9} \rfloor$).
- (b) Test $\beta_w = \beta_m$ with a Wald test using the NW covariance.
- (c) Run Ljung–Box tests with 22 lags on the residuals and on log RV itself, and compare the R^2 with an AR(1).
- (d) Estimate the unrestricted AR(22) and test the 19 HAR restrictions of A5 with a NW Wald test.

B5: log-HAR for SPY with Robust Inference [Solved] (2/2)

Task (continued)

- (e) Estimate HARQ in levels Bollerslev, Patton & Quaedvlieg (2016),
$$RV_{t+1} = \beta_0 + (\beta_d + \beta_Q \sqrt{RQ_t})RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon$$
, on intraday RV and on total variance; then forecast from January 2022 on an expanding window (forecasts outside the in-sample range replaced by the mean) and run a DM test against HAR in levels under QLIKE.
- Report: the coefficient table with both SEs, the Wald, Ljung–Box and AR(22) statistics with p -values, $\hat{\beta}_Q$ with its SE, and the DM result.
- Interpretation: which component matters most? Do robust SEs change any conclusion? Has HAR captured the long memory? Does the measurement-error correction pay off out of sample?

B5: log-HAR for SPY with Robust Inference – Solution [Solved]

	$\hat{\beta}$	SE OLS	SE NW	t_{NW}
β_o	-0.079	0.025	0.027	-3.0
β_d	0.362	0.031	0.037	9.8
β_w	0.337	0.049	0.048	7.0
β_m	0.167	0.046	0.044	3.8

Solution

- (a) All slopes are significant with NW errors ($t = 9.8, 7.0, 3.8$); NW errors are up to 20% larger than OLS ones but no conclusion changes; $R^2 = 0.470$ against 0.413 for an AR(1) ($\hat{\phi} = 0.643$); persistence 0.866.
- (b) Wald $\beta_w = \beta_m$: 4.17, $p = 0.041$: rejected at 5%, the weekly weight is larger.
- (c) Ljung–Box(22): log RV 4641, residuals 17.5 ($p = 0.73$): three coefficients absorb the long memory.

B5: log-HAR for SPY with Robust Inference (extensions) – Solution [Solved]

Solution

- (d) $W = 18.1$, $p = 0.51$ (χ^2_{19}); R^2 0.477 (AR(22)) vs 0.470 (HAR): the 19 restrictions are not rejected, so HAR loses almost nothing and saves 19 parameters.
- (e) $\hat{\beta}_Q = -0.0088$ (SE 0.0042) on intraday RV, significant at 5%; -0.0022 (SE 0.0024) on total variance, not significant.
- (e) Out of sample: QLIKE 0.344 (HARQ) vs 0.326 (HAR), DM $t = 1.23$, $p = 0.22$: no significant difference.
- Reading: the sign predicted by A7 appears, but with 5-minute RV the measurement error is small and the overnight return (not covered by RQ) dominates the noise in total variance, so HARQ brings no forecasting gain here.

B6: Forecasting the Next Week [Proposed]

Solution discussed in class

Task

- Data: SPY total daily variance (as in B5) and SPY daily adjusted closes since 2000 for GARCH; target $Y_t = \sum_{j=0}^4 \text{RV}_{t+j}$, forecast the evening before t , from January 2022 (1176 forecasts).
- (a) Forecast Y_t with a direct log-HAR regression of $\ln Y$ on the HAR regressors, re-estimated daily on an expanding window using only fully observed targets; add $\hat{s}^2/2$ before exponentiating (lognormal approximation; [Duan \(1983\)](#)'s smearing factor is the distribution-free alternative).
- (b) Forecast Y_t with GARCH(1,1)- t (re-estimated every 21 days): $\sum_{h=0}^4 [\bar{\sigma}^2 + (\alpha + \beta)^h (\sigma_t^2 - \bar{\sigma}^2)]$.
- (c) Compute QLIKE and MSE, a DM test on QLIKE with HAC errors, and MZ regressions with the joint Wald test of $(a, b) = (0, 1)$.
- (d) Run the conditional Giacomini–White test [Giacomini & White \(2006\)](#) with instruments $(1, d_{t-5})$; its theory needs fixed-length estimation windows, so re-estimate both methods on rolling windows (log-HAR: last 250 targets; GARCH: last 1000 returns); under the null $Z_t d_{t+5}$ is uncorrelated beyond lag 4, so use the unweighted autocovariances of lags 0–4.
- Report the losses, test statistics and p -values. Interpretation: does the log-HAR advantage survive at a weekly horizon, what do overlapping targets do to the DM test, and which model is biased? Model: B5, A8.

B7: Do Jumps Help? The HAR-CJ Model [Proposed]

Solution discussed in class

Task

- Data: SPY as in B5; continuous part $C_t = RV$ without the significant jumps of B1 (0.1%) plus the squared overnight return (a modelling convention: the overnight move is not split into continuous and jump parts); jump part J_t [Andersen, Bollerslev & Diebold \(2007\)](#).
- (a) Estimate the log specification of [Andersen, Bollerslev & Diebold \(2007\)](#): $\ln$ of total variance on day $t + 1$ on $\ln C_t$, $\ln C_t^{(w)}$, $\ln C_t^{(m)}$ and $\ln(1 + J_t)$, $\ln(1 + J_t^{(w)})$, $\ln(1 + J_t^{(m)})$ (arithmetic 5- and 22-day means, logged afterwards), with Newey–West SEs; test $\beta_{Jd} = \beta_{Jw} = \beta_{Jm} = 0$ at 5%.
- (b) Forecast from January 2022 on an expanding window and compare QLIKE with log-HAR using a DM test.
- Report: the coefficients with SEs, the two R^2 , the two QLIKE values and the DM statistic.
- Interpretation: is the jump coefficient significant, and with what sign did you expect it? Why can a significant coefficient give no forecasting gain? Model: B5, B1.

B8: How Rough Is Volatility? [Proposed]

Solution discussed in class

Task

- Data: SPY daily series 2020–2026 of four measures: total variance, intraday RV, BV and 30-minute subsampled RV; $\ln \sigma_t = \frac{1}{2} \ln(\text{measure})$ [Gatheral, Jaisson & Rosenbaum \(2018\)](#).
- (a) Estimate H from the scaling law $m(q, \Delta) \propto \Delta^{qH}$ with $q \in \{0.5, 1, 1.5, 2, 3\}$ and $\Delta = 1, \dots, 30$ days: slope ζ_q for each q , then H as the slope of ζ_q on q .
- (b) Build a 95% CI for each $\hat{H}$ with a moving-block bootstrap (blocks of 60 days, 300 resamples), computing the increments only within each resampled block, so that no increment spans an artificial join.
- (c) Simulate a Gaussian random walk of the same length (increments $N(0, 0.1^2)$, seed 42), subtract the straight line joining its endpoints (a Brownian bridge, still $H = 0.5$), estimate H , then add i.i.d. $N(0, 0.25^2)$ noise and estimate again; one path, an illustration, not a Monte Carlo study.
- Report the table of $\hat{H}$ [CI] and the two simulated estimates. Interpretation: is H below 0.5 for every measure, and what does measurement error do to $\hat{H}$? Model: B3.

B9: Which Models Survive? The Model Confidence Set [Proposed]

Solution discussed in class

Task

- Data: SPY one-day forecasts of total variance, January 2022 – September 2026 (1 180 days), from log-HAR, HAR, GARCH(1,1)- t , EWMA and yesterday's RV (Lecture 9).
- (a) Add HARQ (B5(e)) and SHAR [Patton & Sheppard \(2015\)](#): replace $\beta_d RV_t$ by $\beta^+ RS_t^+ + \beta^- RS_t^-$, with semivariances $RS_t^\pm = \sum_i r_{t,i}^2 1\{r_{t,i} \gtrless 0\}$ and the overnight return counted as one more return; estimate in levels on an expanding window.
- (b) Compute the mean QLIKE of the seven models.
- (c) Compute the model confidence set at 90% [Hansen, Lunde & Nason \(2011\)](#) with the range statistic and a block bootstrap (blocks of 20 days, 10 000 resamples), and report the MCS p -value of every model.
- Report the table. Interpretation: how many models survive, does the answer change at 75%, and why is the MCS preferable to 21 pairwise DM tests? Model: B5, B6.

C1: Is Bitcoin's Volatility More Predictable Than the S&P 500's? [Proposed]

Solution discussed in class

Task

- Data: the same out-of-sample window for both assets, March 2025 – September 2026: SPY total variance (389 days) and Bitcoin 24-hour RV (422 days), with the log-HAR, GARCH(1,1)- t and yesterday's-RV forecasts of Lecture 9.
- (a) Define “predictable” with three measures and compute each: R^2 of log RV on the log forecast, the QLIKE gain of log-HAR over yesterday's RV, and its gain over GARCH.
- (b) Build a 95% CI for the difference in R^2 (Bitcoin minus SPY) with a moving-block bootstrap in calendar time: the same blocks of 28 calendar days (about 20 SPY trading days) for both assets, 2000 resamples, so their dependence is kept.
- (c) Add a weekend dummy to the Bitcoin log-HAR (what you learned in B2), forecast again, and run a DM test under QLIKE with and without the dummy.
- Report the table, the CI and the DM result, and answer the question in two sentences. Project extension: add Ethereum or a BVB stock, and the [Pele & Mazurencu-Marinescu-Pele \(2019\)](#) entropy measure as a regressor.

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have SPY trade prices, every tick from 09:30 to 16:00, in a pandas Series px with a time index. Compute the daily realised variance, the annualised volatility and the HAR regressors."

The AI answer: text

- $RV_t = \sum_i r_{t,i}^2$, with $r_{t,i}$ the intraday log returns of day t [Andersen et al. \(2001\)](#)
- Use every tick: more returns always make RV_t more precise
- RV_t is SPY's daily variance: compare it directly with squared close-to-close returns and with the GARCH variance
- HAR: regress RV_{t+1} on RV_t and on its 5-day and 22-day averages [Corsi \(2009\)](#)

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np, pandas as pd
lp = np.log(px)
r = lp.groupby(lp.index.date).diff().dropna()
RV = (r**2).groupby(r.index.date).sum()
vol_ann = np.sqrt(RV) * 252
H = pd.DataFrame({'d': RV,
                  'w': RV.rolling(5).mean(),
                  'm': RV.rolling(22).mean(),
                  'y': RV.shift(-1)}).dropna()
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ What does A3 say about $\mathbb{E}[RV^{(n)}]$ when prices contain noise?
- ▣ What happens to SPY between 16:00 and 09:30 the next morning?
- ▣ Plug in a typical day, $RV_t = 1.0 \times 10^{-4}$ (1%²): is the annualised volatility plausible for SPY?

C3: Does Machine Learning Beat HAR for SPY? [Proposed] (1/3)

Solution discussed in class

Task

- Context: Christensen, Siggaard & Veliyev (2023) find that random forests and neural networks beat HAR once predictors beyond past RV enter, and that the gain grows with the horizon (Lecture 9, case study). You test this on SPY with their sample split and their model settings.
- (a) Build the daily open-to-close RV of SPY from the 5-minute bars of the regular session (78 returns a day, as in B1), and three targets: RV_{t+1} , the average RV over the next 5 days and the average over the next 22 days (Section 5 of the paper).
- (b) Build two information sets dated t : $\mathcal{M}_{\text{HAR}}$ with the daily, weekly and monthly RV, and $\mathcal{M}_{\text{ALL}}$, which adds the VIX, the squared daily log return of the Hang Seng (both in data/market), the 1-week SPY momentum, the first difference of log dollar volume, the first difference of the 3-month T-bill rate (FRED, DTB3), the daily economic policy uncertainty index (FRED, USEPUINXD) and the Aruoba–Diebold–Scotti business-conditions index (Federal Reserve Bank of Philadelphia). Date each predictor by its release, not its reference date: use only values published by the close of day t (for the Aruoba–Diebold–Scotti index, which is revised, the Philadelphia Fed’s real-time vintages) and disclose any series for which vintages are unavailable.
- State which of the paper’s predictors (Table 1) cannot be built for an index fund, and why: this is a data limitation, not a reason to change the method.

C3: Does Machine Learning Beat HAR for SPY? [Proposed] (2/3)

Solution discussed in class

Task (continued)

- (c) Split the sample as in Appendix A.1 of the paper: the first 1000 usable days for training, the next 200 for validation, the rest for the test; a row belongs to a subset only if its whole target window lies inside it (drop the last $h - 1$ rows before each boundary), and report the usable sample size per horizon; standardise every predictor with the training mean and variance only.
- (d) Estimate HAR, log-HAR (forecast $\exp(\hat{f} + \hat{\sigma}^2/2)$) and OLS on all predictors on a rolling window over training and validation merged; the elastic net on a rolling training window, without merging, with λ and the mixing weight chosen on the validation set (Section 3); a random forest with 500 trees on the merged window, $\lfloor J/3 \rfloor$ candidate predictors per split (J : number of predictors) and minimum node size 5 (Table 15); and NN_2^{10} : layers of 4 and 2 neurons, leaky ReLU, Adam with learning rate 0.001, dropout 0.8, early stopping with patience 100, the average of the best 10 of 100 seeds, trained once on a fixed window (other settings as in Table 15). HAR keeps its three RV regressors under both information sets. Replace negative forecasts with the smallest in-sample RV.

C3: Does Machine Learning Beat HAR for SPY? [Proposed] (3/3)

Solution discussed in class

Task (continued)

- (e) Report a table of test-set MSE and QLIKE relative to HAR for the three horizons and both information sets, the one-sided DM p -values against HAR (HAC errors for the overlapping 5- and 22-day targets, with a bandwidth above $h - 1$, since Bartlett weights shrink lag $h - 1$, and a sensitivity check over bandwidths) and the 90% MCS, as in B9.
- Interpretation: does the ranking of Tables 2, 3 and 7 of the paper survive on SPY, does the ML advantage grow with the horizon, and does it survive QLIKE? With a test set of this length, what can a non-significant DM test tell you? Model: B5, A6.
- Project extension: repeat on Bitcoin's 24-hour RV and compare the accumulated local effects of RV and VIX across models (Figure 7 of the paper).

Common Pitfalls

- ▣ Sampling every tick without checking the signature plot
- ▣ Forgetting the overnight return: intraday RV understates the daily variance of SPY by about a third
- ▣ Treating RV as the true variance: it has a standard error, and it makes R^2 of forecasts look low
- ▣ Comparing volatility forecasts with MAE or with MSE on volatility: not robust to the noisy proxy
- ▣ Usual OLS standard errors in HAR and i.i.d. bootstrap for daily RV: both ignore autocorrelation
- ▣ Reading every significant jump test as a news jump: thousands of tests give false alarms

Take-Home Assignment (Team Project)

- Choose an asset with intraday data not covered in class (Ethereum, a currency pair, a stock)
- Build daily RV, BV and a jump component; show the signature plot and the intraday pattern
- Compare log-HAR (with at least one extension) and GARCH out of sample with QLIKE, DM and MZ
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Nonparametric and Semiparametric Models
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

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